CO2 - ASSESMENT TOOL 2

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COURSE CODE:	**ITA0329**
COURSE:	**MOBILE COMPUTING**

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication

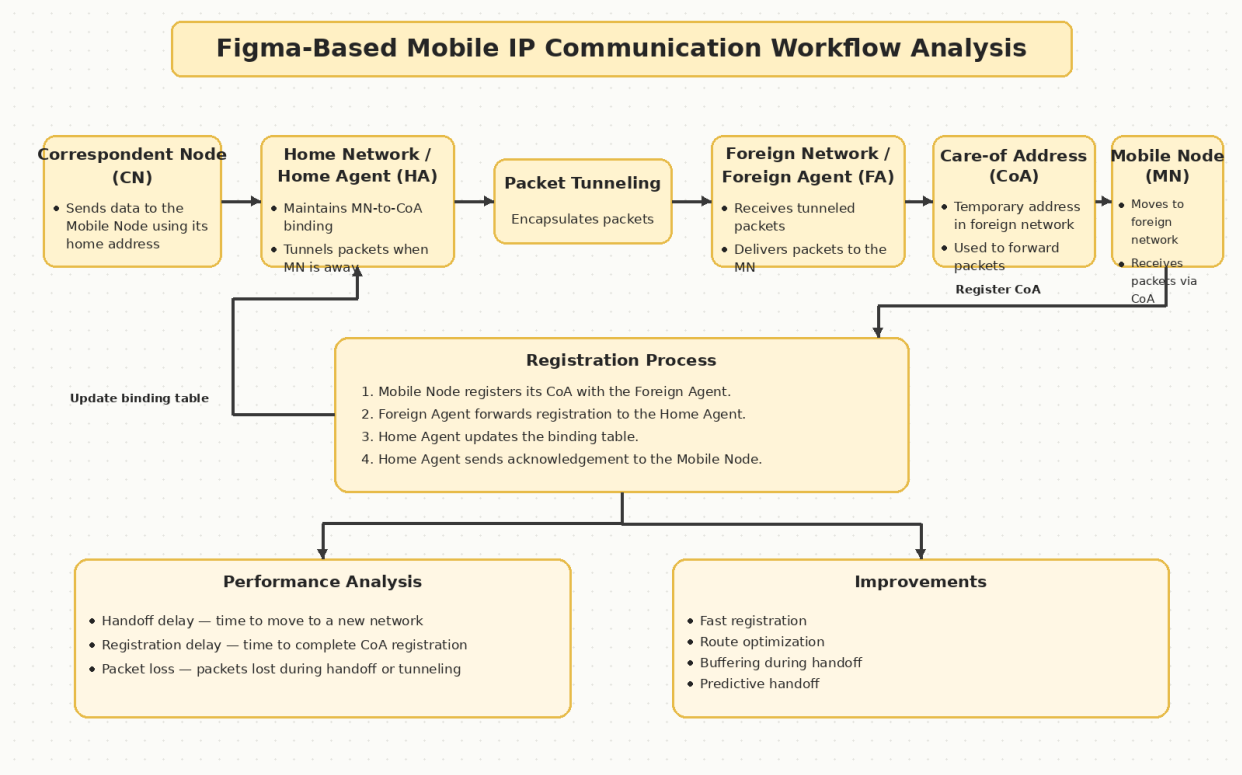
when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include

the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA),

Registration Process, and Packet Tunneling.

Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.



3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

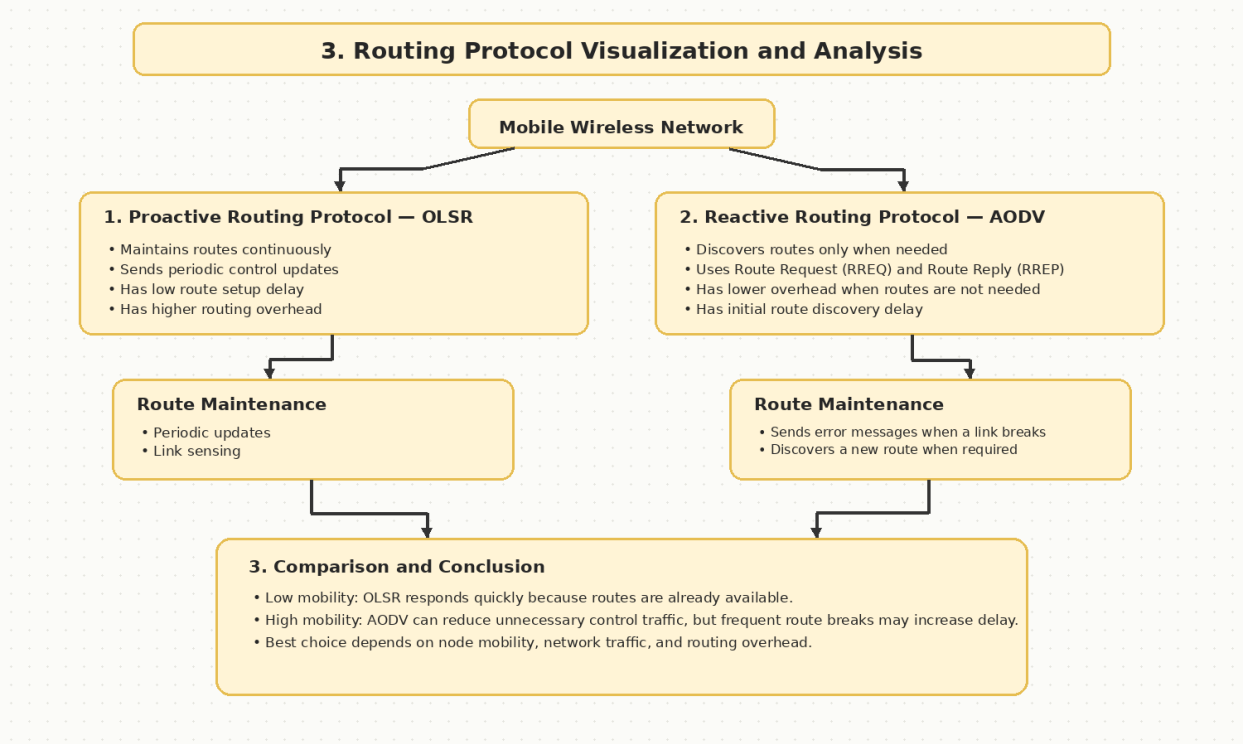
Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework

illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.


3. Routing Protocol Visualization and Analysis
Mobile Wireless Network
1. Proactive Routing Protocol — OLSR
• Maintains routes continuously
• Sends periodic control updates
• Has low route setup delay
• Has higher routing overhead
2. Reactive Routing Protocol — AODV
• Discovers routes only when needed
• Uses Route Request (RREQ) and Route Reply (RREP)
• Has lower overhead when routes are not needed
• Has initial route discovery delay
Route Maintenance
• Periodic updates
• Link sensing
Route Maintenance
• Sends error messages when a link breaks
• Discovers a new route when required
3. Comparison and Conclusion
• Low mobility: OLSR responds quickly because routes are already available.
• High mobility: AODV can reduce unnecessary control traffic, but frequent route breaks may increase delay.
• Best choice depends on node mobility, network traffic, and routing overhead.